Kunhee (Geoffrey) Kim

DATA ENGINEER · DATA ANALYST

kunheekim02@gmail.com | Berkeley / Bay Area, CA | linkedin.com/in/kk02 | github.com/kheekim02

SUMMARY

Data Engineer and UC Berkeley Data Science graduate who designs and scales production ETL/ELT pipelines, multi-terabyte data lakes and warehouses (PostgreSQL, Neo4j, Apache Spark), and ML/NLP systems. Scaled a research data lake to 2M+ filings and 33M+ records with strict data-integrity guarantees, relational data modeling, and SQL performance tuning. Bilingual in English and Korean.

EXPERIENCE

Financial Data Analyst — Global Key Advisors, Bay Area, CA *January 2026 – Present*

- Re-architected SEC EDGAR extract-transform-load (ETL) scraping pipelines (8-K, 10-K, 10-Q, Form 15/25) from a memory-constrained CSV approach to a threaded PostgreSQL integration, scaling the data lake to **2.05M+ text filings and 3.3M+ financial records across 28,000+ entities**.
- Administered a PostgreSQL research warehouse (*sec_data*) holding **33M+ records and 42 GB of text**; architected modular schemas, tuned TOAST storage, and debugged the query planner via system catalogs and ANALYZE statistics.
- Automated daily ingestion of a 1.12 GB price panel from a remote Windows host to a Linux Spark cluster, **eliminating IP-ban risk and saving hours/week**; built data-integrity logic (CIK joins, point-in-time mapping, vendor dedup, idempotent upserts) to prevent silent corruption.
- Built NLP extraction with local LLMs (Ollama) parsing filings into corporate-relationship edges in a **Neo4j knowledge graph** via an async pipeline processing **~480 filings/hour** unattended.
- Migrated prediction strategy from FinBERT sentiment to Graph Neural Networks (GNN); out-of-sample backtesting **exposed lookahead bias** (a perceived +2.39% return collapsed to negative under 30-bps costs), preventing a losing strategy deployment.

PROJECTS

Alpha Signals — Quantitative SEC Event Research Pipeline *Jan 2026 – Present*

Python, Polars, PostgreSQL, Neo4j, Ollama, Graph Neural Networks

- Flagship pipeline testing whether corporate events in SEC filings predict stock returns; built an autonomous daily loop (signal generation to simulated trades to GNN predictions by 04:30 PDT) over a Neo4j knowledge graph spanning 28,000+ entities.

PathWise — Intelligent iOS Running App *Jan 2026 – May 2026*

Swift (SwiftUI, MapKit, ActivityKit), FastAPI, PostgreSQL + PostGIS, GraphHopper, Docker

- Full-stack architect: built a pathfinding orchestrator integrating GraphHopper with PostGIS for real-time geofenced routing and a native iOS client with turn-by-turn voice navigation; achieved **sub-2-second route generation**. Presented at Berkeley's startup pitch competition.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Swift, Bash/Shell

Data Engineering: ETL/ELT pipeline design, data lake & warehouse architecture, relational data modeling, schema design, idempotent ingestion, data quality & validation, deduplication, web scraping

Databases & Big Data: PostgreSQL (performance tuning, query optimization), Neo4j, Apache Spark (PySpark), PostGIS, SQLite

Python & Backend: Pandas, Polars, FastAPI, SQLAlchemy, Pydantic, concurrent.futures, BeautifulSoup, REST APIs

ML / NLP: Local LLM inference (Ollama), Graph Neural Networks (GNN), entity resolution, backtesting, event studies

Tools & Workflow: Docker, Linux, Git, cron/systemd orchestration, pg_dump/pg_restore, Tableau

Languages (Spoken): English (fluent), Korean (fluent)

EDUCATION

University of California, Berkeley — B.A. Data Science *May 2026*

GPA 3.81 · Completed in 3 years · 2-year military service in Korea (2022–2024)

Certificates: Berkeley Changemaker; Entrepreneurship & Technology (Sutardja Center / SCET, UC Berkeley)

RESEARCH & LEADERSHIP

- **Student Researcher, UC Berkeley** (Prof. Jill Berrick, *FamiliesNotFees*) — Applied statistical modeling and ML to national child-welfare data; built Tableau dashboards and the project website, growing reach to 4,000+ views / 1,000+ unique visitors.
- **Course Coordinator, ENGIN 270B** (2025) — Ran operations for a graduate course of 100+ students; built automated Excel tracking surfacing data-driven metrics for curriculum improvement.
- **Berkeley Data Discovery** — Meta-analysis of national transfer metrics using ML on large-scale education datasets, with interactive dashboards for strategic decision-making.